



# Cell Injury

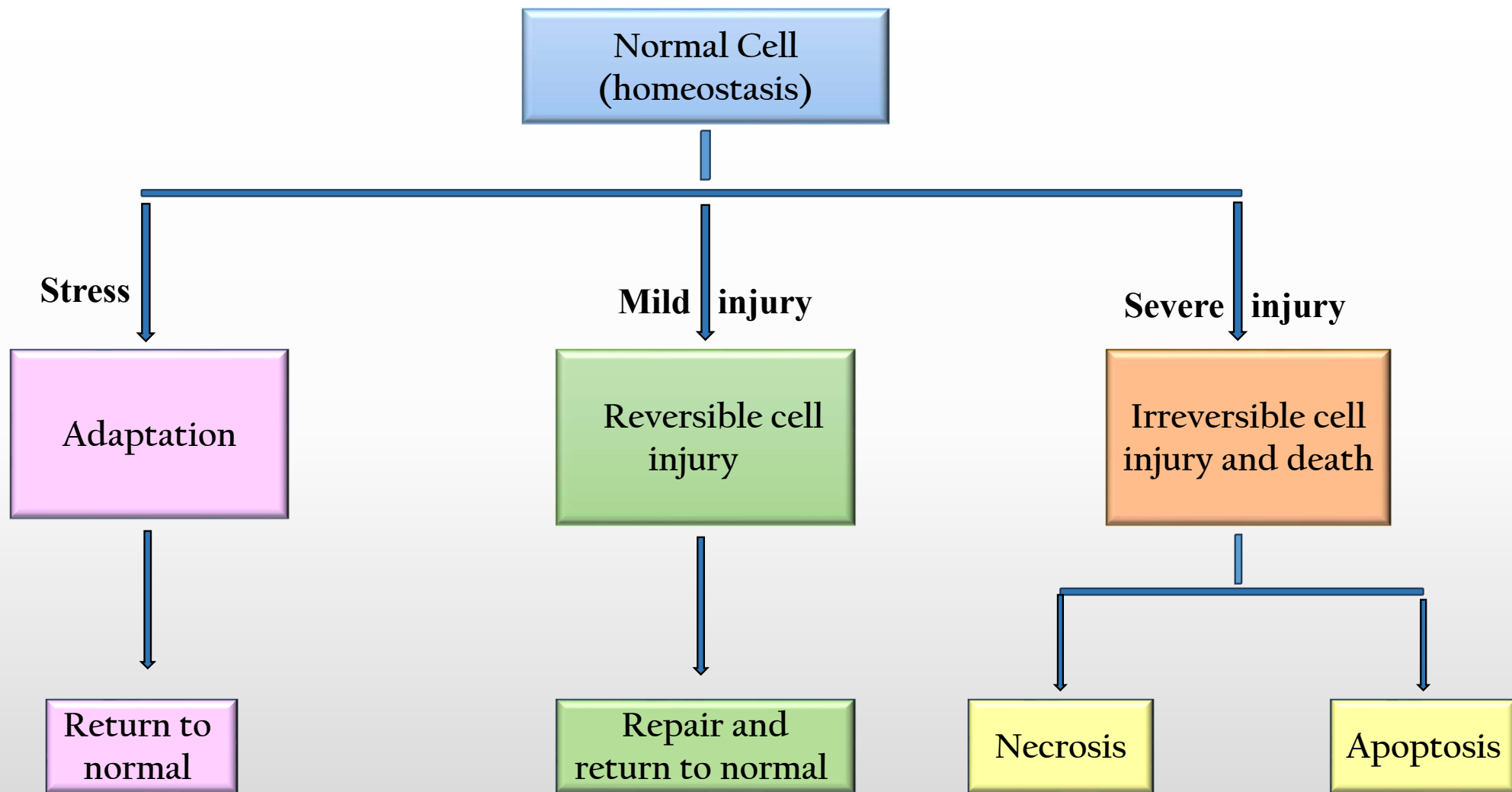
## Lec. 2

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# Cellular responses to stress and noxious stimuli

Cells actively interact with their environment, constantly adjusting their structure and function to accommodate changing demands and extracellular stresses. The intracellular milieu of cells is normally tightly regulated such that it remains constant, a state referred to as **homeostasis**.

As cells encounter physiologic stresses or potentially injurious conditions ,they can undergo adaptation . If the adaptive capability is exceeded or if the external stress is harmful or excessive, cell injury develops .

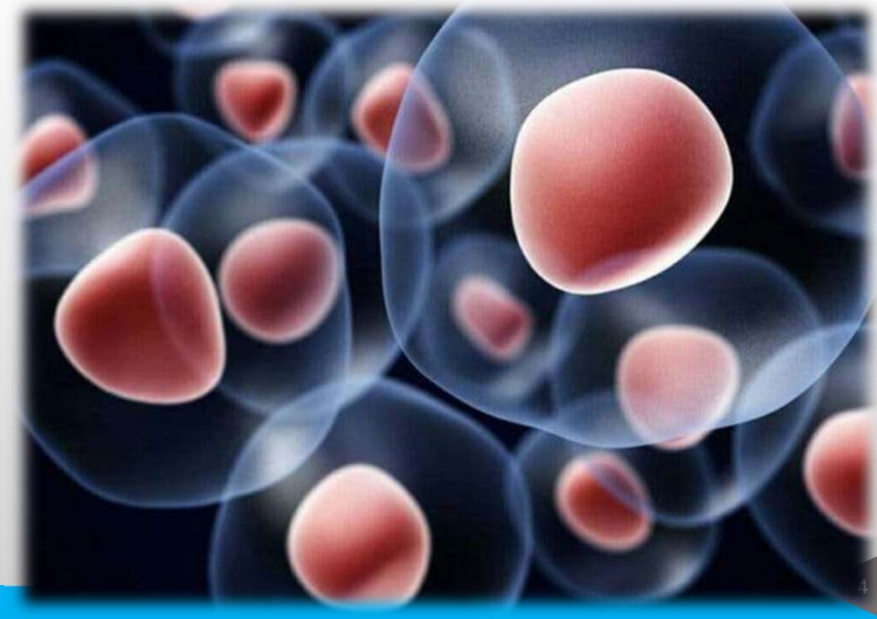


# Definition of cell injury

**Cell injury** : the sequence of changes that occur in the living cells after exposure to injurious stimuli (causes) .

**The extent of cell injury depends on :**

- 1) Type , duration and severity of the cause .
- 2) Cell type , status and adaptability .



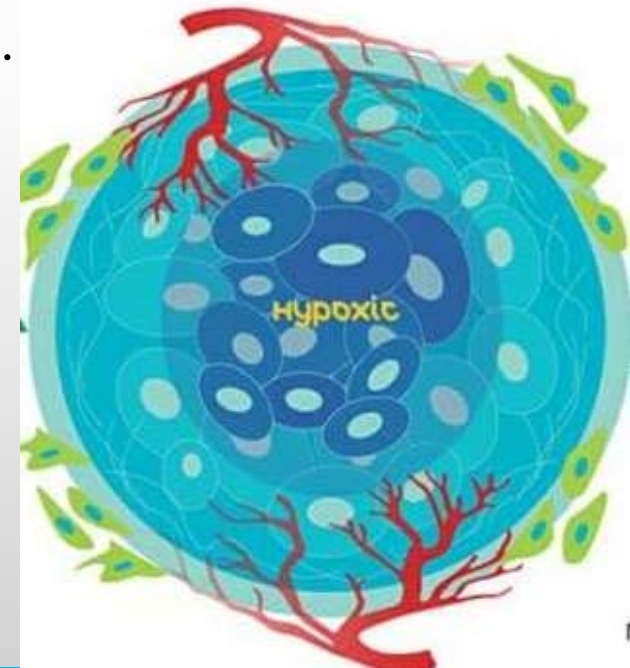
# Causes of cell injury

## 1) Hypoxia and ischemia

Hypoxia, which refers to oxygen deficiency, and ischemia, which means reduced blood supply, are among the most common causes of cell injury .

## 2) Toxins

Include air pollutants, insecticides, CO, cigarette smoke, ethanol, and drugs. Many drugs in therapeutic doses can cause cell or tissue injury in a susceptible patient or in many individuals if used excessively or inappropriately .



### 3) Infectious agents

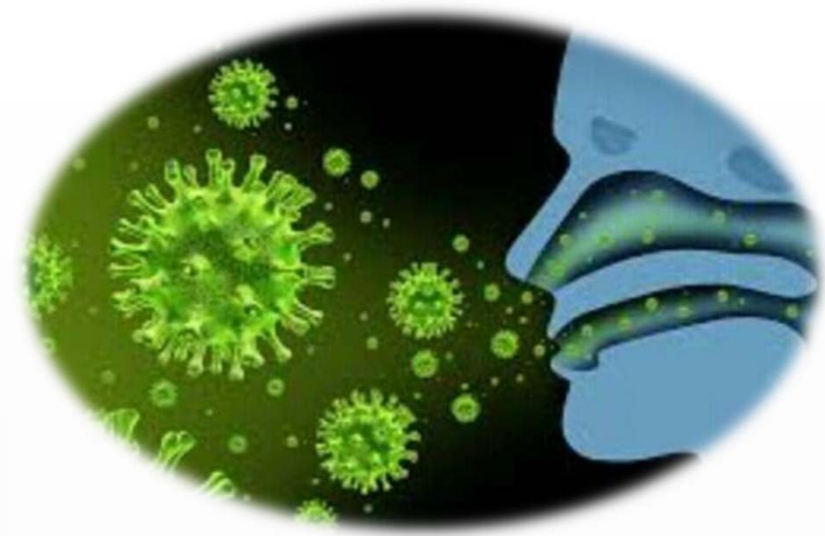
including viruses, bacteria, fungi and protozoans .

### 4) Physical agents

Such as trauma, extremes of temperature, radiation and electric shock .

### 5) Immunologic reactions

Although the immune system defends the body against pathogenic microbes, immune reactions can also result in cell and tissue injury like autoimmune reactions against one's own tissues and allergic reactions against environmental substances.



## **6 ) Genetic abnormalities**

Genetic defects can result in pathologic changes like congenital malformations associated with Down syndrome and sickle cell anemia.

## **7 ) Nutritional imbalances**

Like dietary insufficiency and dietary excesses .

## **8 ) Aging**

Cellular senescence leads to alterations in replicative and repair abilities of individual cells and tissues.

# Types of cell injury

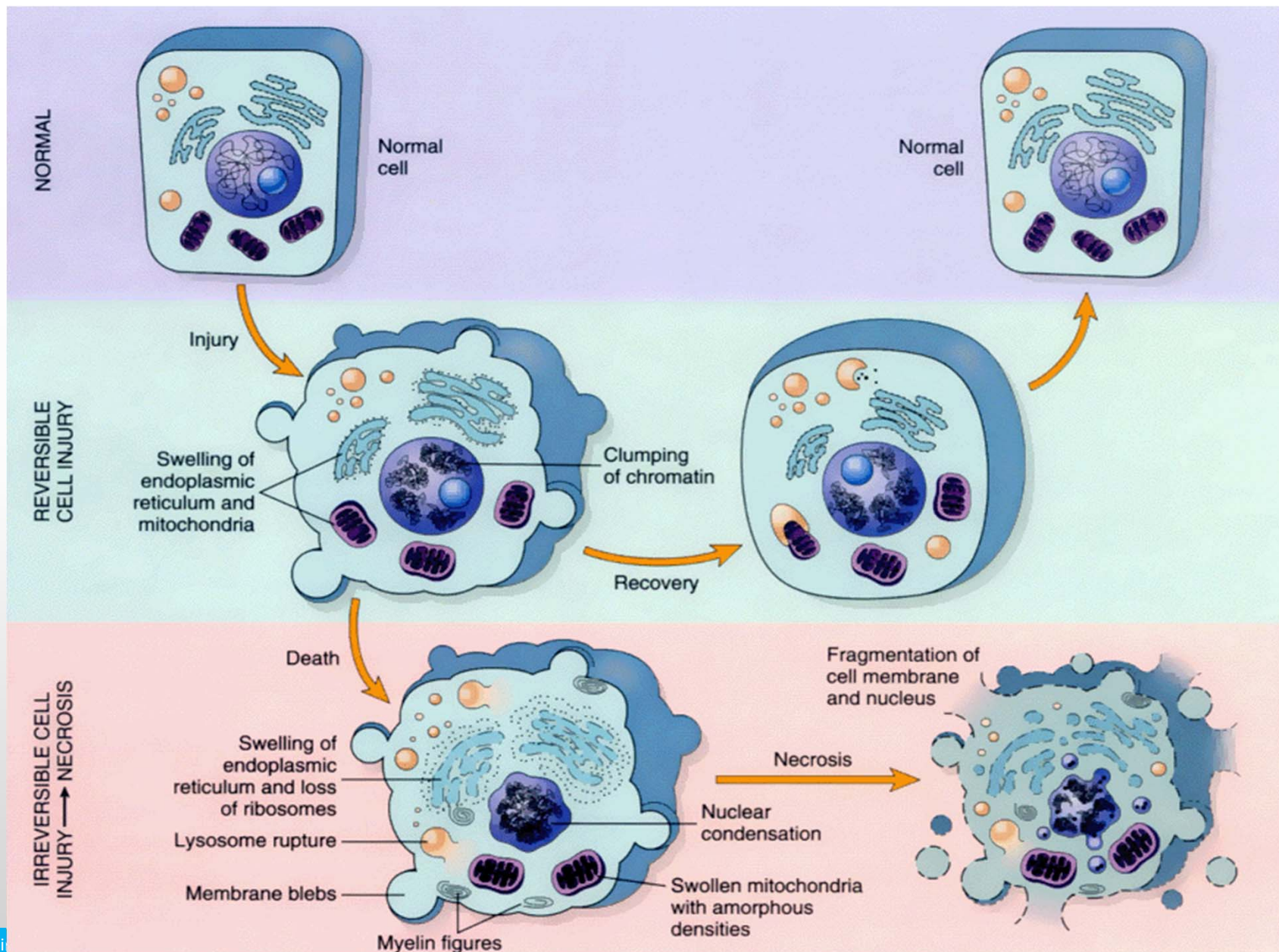
## 1) Reversible cell injury

is the stage of cell injury at which the deranged function and morphology of the injured cells can return to normal if the damaging stimulus is removed .

## 2) Irreversible cell injury and cell death

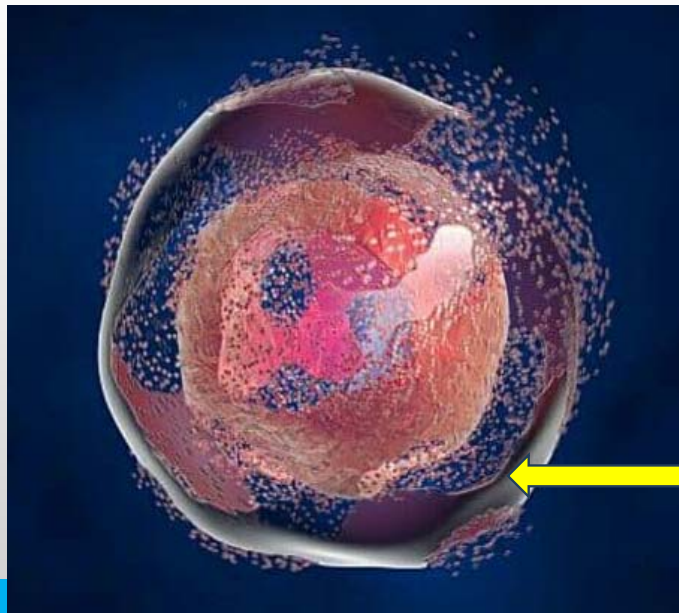
When the stress is severe and progressive , the injury becomes irreversible , the cell is unable to revert back and end up in permanent damage ( cell death ).





# Types of cell death

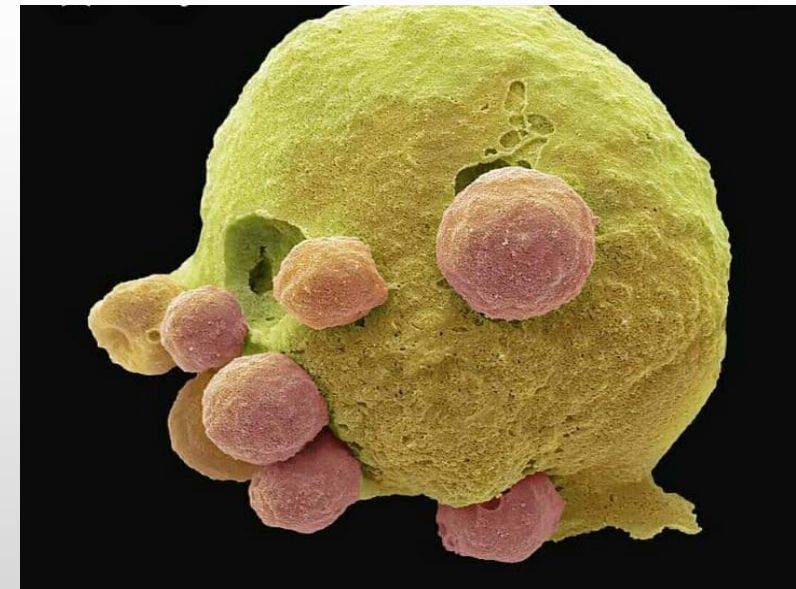
- 1) **Necrosis** : is a form of cell death in which cellular membranes fall apart, and cellular enzymes leak out and ultimately digest the cell . Necrosis elicits a local host reaction, called inflammation, that is induced by substances released from dead cells.



**Necrotic cell**

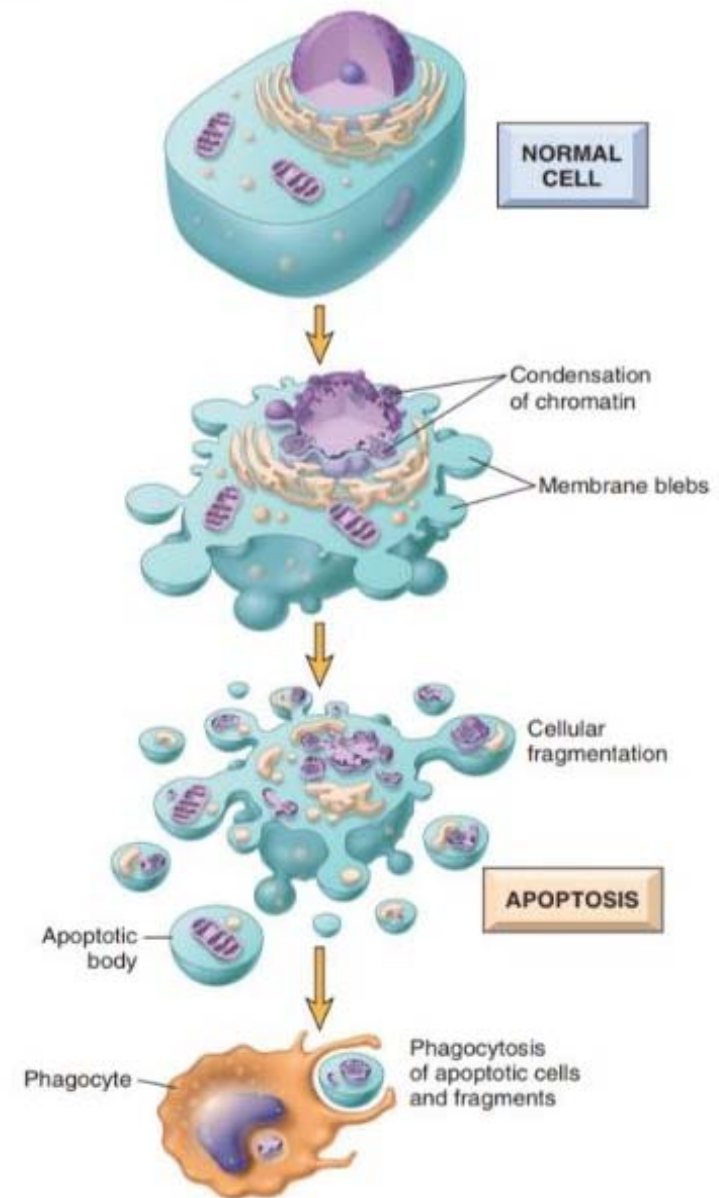
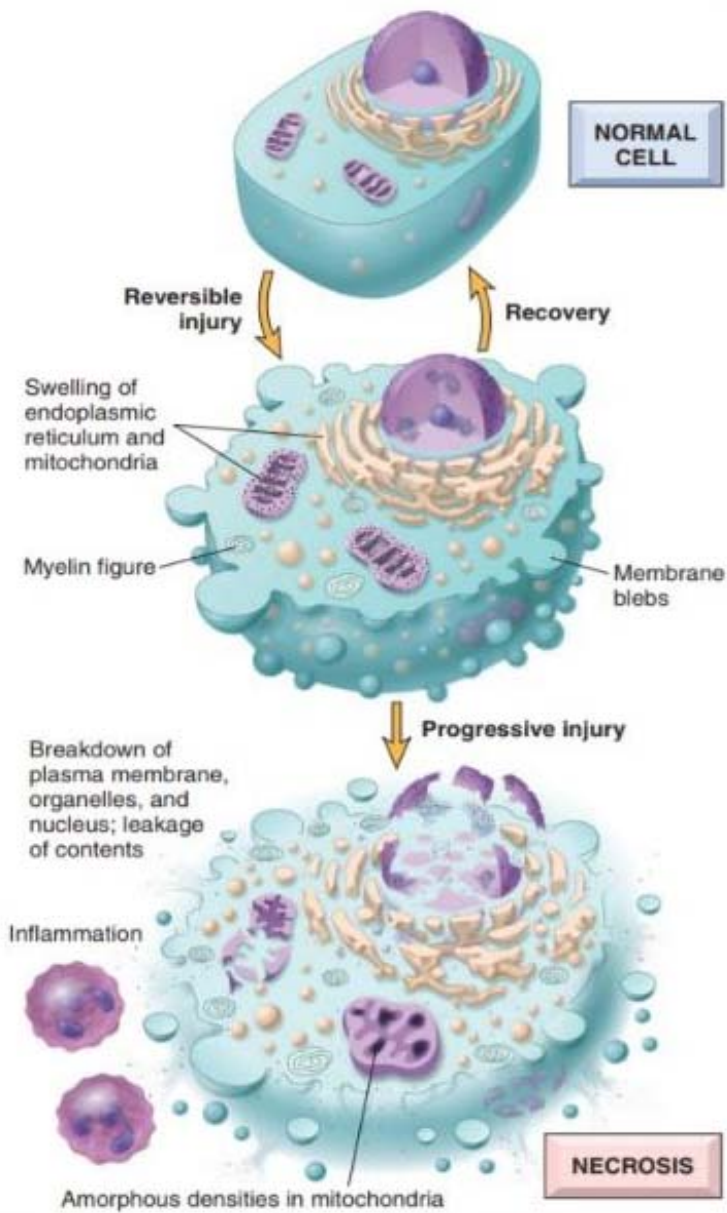
- **2) Apoptosis** : regulated mechanism of cell death that serves to eliminate unwanted and irreparably damaged cells, with the least possible host reaction . In apoptosis , the cells activate enzymes that degrade the cells' own nuclear DNA and nuclear and cytoplasmic proteins without complete loss of membrane integrity.

*Whereas necrosis is always a pathologic process, apoptosis serves many normal functions and is not necessarily associated with pathologic cell injury.*



***Apoptotic cell***

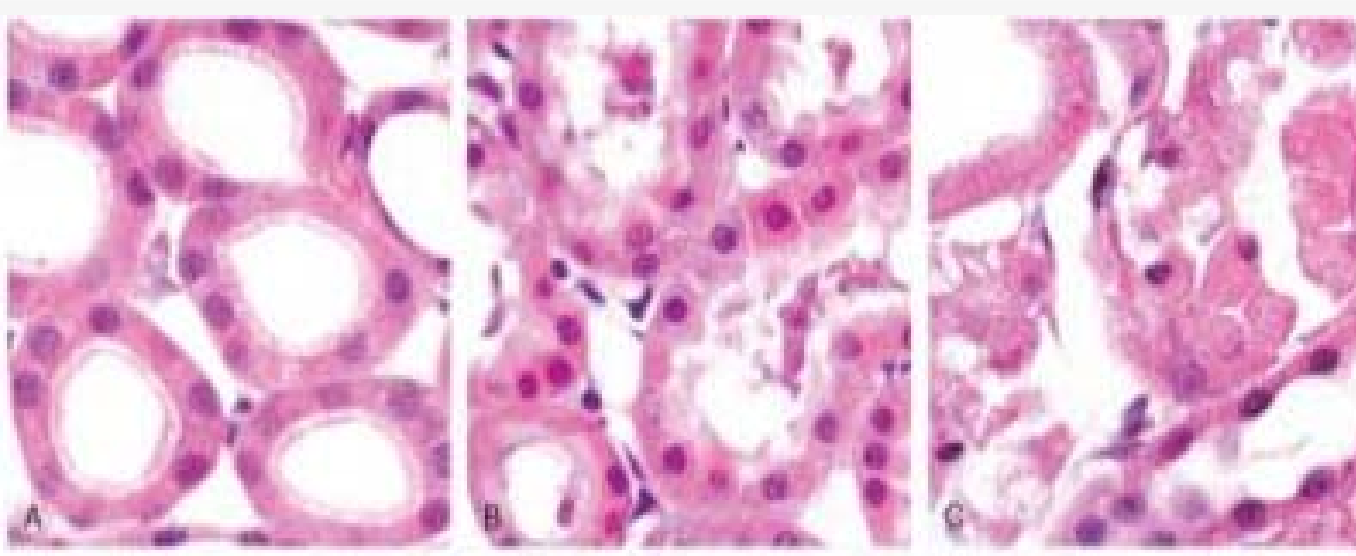




| Feature                              | Necrosis                                                                 | Apoptosis                                                                                                                                                        |
|--------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cell size                            | Enlarged (swelling)                                                      | Reduced (shrinkage)                                                                                                                                              |
| Nucleus                              | Pyknosis →<br>karyorrhexis →<br>karyolysis                               | Fragmentation into<br>nucleosome-sized<br>fragments                                                                                                              |
| Plasma<br>membrane                   | Disrupted                                                                | Intact; altered structure,<br>especially orientation<br>of lipids                                                                                                |
| Cellular contents                    | Enzymatic digestion;<br>may leak out of cell                             | Intact; may be released<br>in apoptotic bodies                                                                                                                   |
| Adjacent<br>inflammation             | Frequent                                                                 | No                                                                                                                                                               |
| Physiologic or<br>pathologic<br>role | Invariably pathologic<br>(culmination of<br>irreversible cell<br>injury) | Often physiologic<br>means of eliminating<br>unwanted cells; may<br>be pathologic after<br>some forms of cell<br>injury, especially<br>DNA and protein<br>damage |

# Morphology of cell injury

- **Morphology of reversible injury:** Cell swelling, fatty change, plasma membrane blebbing and loss of microvilli, mitochondrial swelling, dilation of the ER, eosinophilia.



(A) Normal kidney tubules with viable epithelial cells. (B) Early(reversible)ischemic injury showing surface blebs, increased eosinophilia of cytoplasm, and swelling of occasional cells.(C) Necrotic(irreversible) injury of epithelial cells, with loss of nuclei and fragmentation of cells.

- **Morphology of irreversible Injury**

- 1) Necrosis :**

- ❖ Chromatin clump , mitochondria swell and rupture of membrane , cell contents spill, inflammatory response triggered .

- 2) Apoptosis :**

- ❖ DNA cleaved at specific sites , cytoplasm shrinks without membrane rupture , blebbing of plasma and nuclear membranes , cell contents in membrane bounded bodies and no inflammation .

- **The most important targets of injurious stimuli are:**

- (1) Mitochondria, the sites of ATP generation (the energy store of cells).
- (2) Cell membranes, on which the ionic and osmotic homeostasis of the cell and its organelles depends.
- (3) Protein synthesis.
- (4) The cytoskeleton.
- (5) The genetic apparatus of the cell.



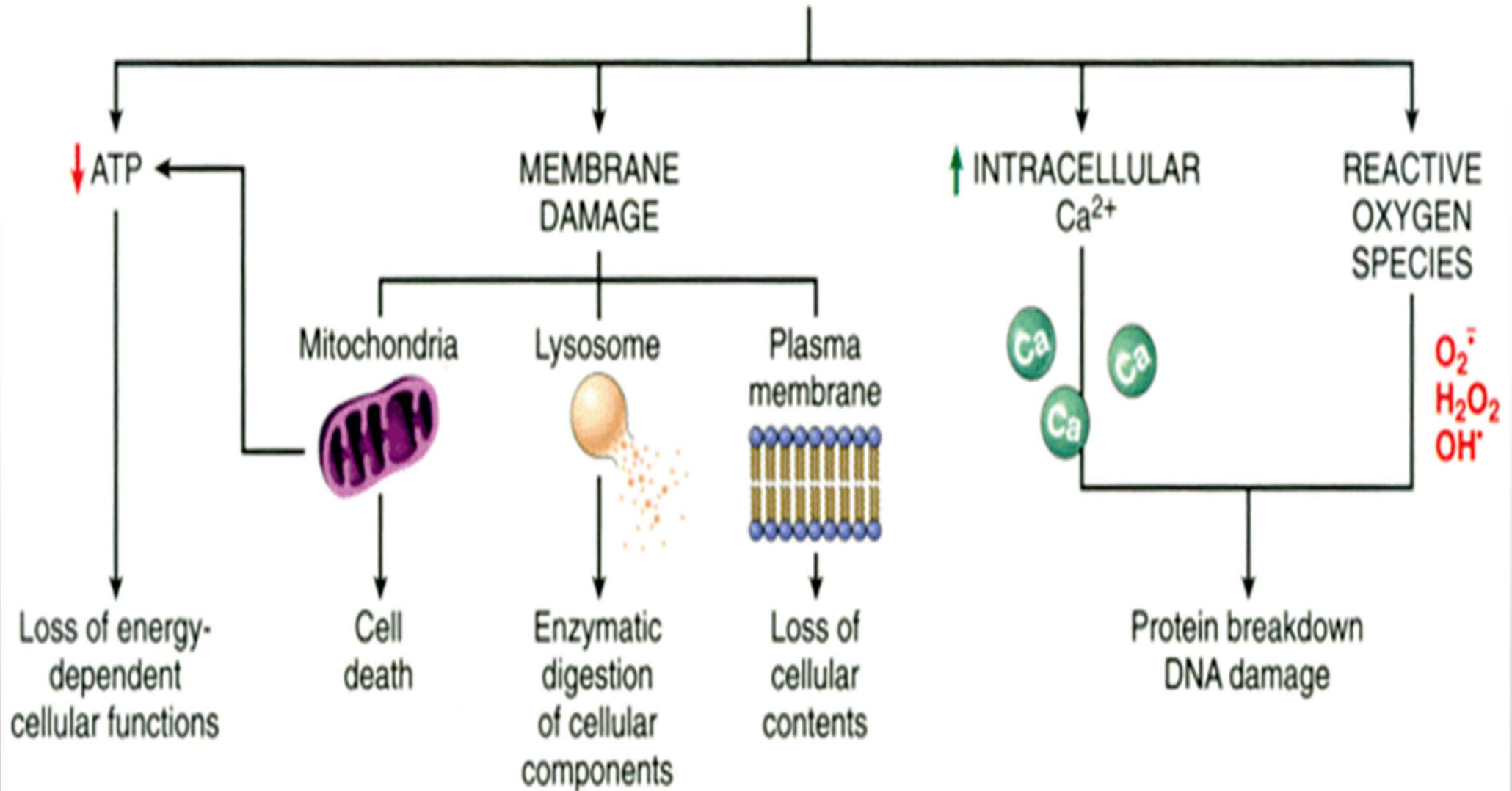
# Mechanisms of cell injury

- 1) **ATP depletion:** The major causes of adenosine triphosphate (ATP) depletion are reduced supply of oxygen and nutrients, mitochondrial damage, and the actions of some toxins (e.g., cyanide).
- 2) **Mitochondrial damage:** ATP depletion → failure of energy-dependent cellular functions.
- 3) **Influx of calcium:** lead to activation of enzymes that damage cellular components and may also trigger apoptosis.

**4) Accumulation of reactive oxygen species:** covalent modification of cellular proteins, lipids, nucleic acids.

**5) Increased permeability of cellular membranes:** the activity of the plasma membrane energy-dependent sodium pump is reduced, resulting in intracellular accumulation of sodium and efflux of potassium, causing cell swelling.

## INJURIOUS STIMULUS



Thank You...